

Bitcoin & Quantum Computing

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Bitcoin & Quantum Computing

- ✦ Signature and Mining overview
- ✦ Bitcoin threat model
- ✦ Timeline for Solution



Quantum computing and Bitcoin

- ✧ Mining
 - ✧ Quantum miners running Grover's Algorithm efficiently
- ✧ Signatures
 - ✧ Cryptographically Relevant Quantum Computers ,CRQCs, breaking cryptographic assumptions (ECC)

Mining

Mining goal: Find a nonce that hashes below a target

Classical mining: More attempts = higher chance

- Linear growth, parallelizable

Quantum mining: Longer runtime = higher chance

- Square-root speedup, not parallelizable

Bitcoin Threat Model - Mining

- ✦ Mining will be significantly less affected in the foreseeable future
- ✦ If quantum mining will become realistic, we can expect:
 - ✦ Significant mining centralization
 - ✦ Increase in stale blocks
 - ✦ Weakening of security assumptions (51% attack)

Asymmetric Cryptography

- ✦ There are two keys - Public Key (**PK**) and Secret Key (**SK**)
- ✦ A **SK** can be used to sign a message
- ✦ The base assumption are:
 - ✦ Calculating **PK** from **SK** - **Easy**
 - ✦ Calculating **SK** from **PK** - **Infeasible**
 - ✦ Forging a signature having only **PK** - **Infeasible**

What would CRQC break?

- ✦ **EVERYTHING!** (Well, almost)
- ✦ Calculating **SK** from **PK** - **Easy**
- ✦ Forging a signature having only **PK** - **Easy**
- ✦ The attacker needs access to the **PK**

Cryptography and Bitcoin

- ✦ For a bitcoin transaction to be valid it needs to be signed by the **SK** matching the **PK** of the utxo
- ✦ In some cases the **PK** is present on-chain
- ✦ In others a **hash** of the **PK** is on-chain, and needs to be revealed together with a signature to move the funds

Bitcoin Threat Model - Signatures

- ✦ Some utxo are **immediately** vulnerable (long-range attack)
 - ✦ **P2PK, P2MS, P2TR**
 - ✦ **Address re-use**
- ✦ All other will be vulnerable **upon spend** (short-range attack)
- ✦ Quantum secure signatures are known, but **lack efficiency or capabilities**
- ✦ There is **active ongoing research** in the cryptography community

UTXO migration

- ✦ ~190 million UTXOs, all need to move
- ✦ ~60 million UTXOs exposed PK on chain
- ✦ Migration estimates vary from ~3 months to ~3 years
- ✦ Given a standard, migration can happen organically

Burn vs. Steal Dilemma

- ✦ CRQC create a security bug
- ✦ What do we do with coins that did not migrate?
- ✦ Burn - coins can never move
 - ✦ Philosophical clash
- ✦ Steal - Assets claimed by those with a CRQC
 - ✦ Destabilize the ecosystem

Timeline for upgrade

Three main phases:

1. Research

2. Implementation

3. Migration

Long-Term

2.5 Y

1.5 Y

3 Y

~7 Y

Contingency

0.5 Y

1 Y

1 Y

~2 Y

Ecosystem Vulnerabilities

- ✦ Bitcoins lives on the [internet](#) (SSL, DNS, etc.)
- ✦ [P2P](#)-level attacks
- ✦ Signatures on [Core versions](#)

Bitcoin & Quantum Computing

- ✧ Bitcoin threat model
 - ✧ Mining
 - ✧ Signatures
- ✧ Burn vs. Steal
- ✧ Timeline to solution



Read the
complete report
here:

[chaincode.com/
bitcoin-post-
quantum.pdf](https://chaincode.com/bitcoin-post-quantum.pdf)

